DTU LiPo Battery Charger v1.0 User Guide

Battery compatibility

The LiPo battery charger is only intended to charge a single LiPo or Li-Ion battery cell at a time. Charging more cells either in series or parallel is not supported. Only LiPo or Li-Ion battery cells are suitable for this product, and other chemistries such as LiFePO4 are not supported as the battery charger will overcharge the LiFePO4 battery.

Power

The type-C USB connector in the corner of the board labeled with "USB" can be used to power the battery charger. Alternatively, the pin header above the USB connector can power the charger with a 5 V power connection. The status LED will indicate when a connected battery is charging. The LED is off when: charging is complete, charger is in shutdown mode or there is no battery connected.

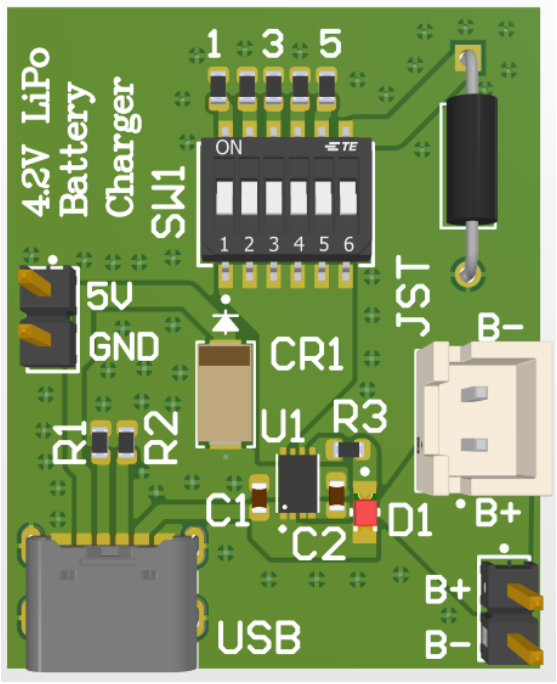
Charge rate selection

Five resistors are fitted to the board, which will decide the rate at which the connected battery is charging. With the six-position dip switch the resistors can be connected to the charger IC. Only one switch should be on at a time. From switch position 1 to 5 the charge rates are: 100, 196, 303, 416 and 500 mA. The final switch controls a user solderable resistor. Use the following equation to find the resistor you need.

$$I_{REG} = \frac{1\ \text{kV}}{R_{PROG}}$$

Battery charger use

To charge a LiPo or Li-Ion battery the battery must be connected first. Use either the pin header or the JST battery connector (model: B2B-HX-A). Then the desired charge rate is chosen by flipping the corresponding dip switch. Only one switch can be on at a time. Lastly the Power is connected to the battery charger, through either the USB connector or the pin header. To ensure that the battery is charging, check that the status LED is lit. If it is off, the connected battery may be fully charged already, damaged or the power connection is faulty.



Safety warnings

Incorrect DIP Switch settings: Using multiple switches at once will change the charge rate. Increasing it as the resistors are connected in parallel.

Connecting multiple batteries: Use only one at a time. Connecting a battery to both connectors results in the batteries being connected in parallel, which is not supported by this product.

Battery Chemistry Compatibility: The charger is only compatible with LiPo batteries. The charger might overcharge other types such as LiFePO4 batteries.

High C-Rating: Choosing a high charge rate may cause damage and shorten lifespan of batteries. Choose a charge rate that corresponds to a rate of 1 C and a max of 2 C. This ensure no damage to the battery and no shortening of its lifespan.